

ALGORITHMS | LEVEL 2 | FINAL PROJECT IDEAS

Final Project Ideas

(Topic #1): Hospital Management System

Hospital Management System: The required data is **Hospital** (Name & Location), **Doctor** (ID & Name & Phone & Specialization), **Patient** (ID & Name & Phone & age).

- 1- Design a java program that takes the required data from user.
- 2- Sort Hospital by names.
- 3- Sort Doctors by its IDs.
- 4- Sort Patients by its IDs.
- 5- Search for all doctors who have the same specialty.
- 6- Search for patient by ID.

Use three different sorting algorithms. Then calculate Big (O) of your program.

Topic (2): Hotel Management System

Hotel Management System: The required data is **Hotel** (Name & Location), **Rooms** (Number & Floor & View & Price & Availability), **Guest** (Number & Name & Phone & SSD).

- 1- Design a java program that takes the required data from user.
- 2- Sort Hotels by their names.
- 3- Sort Rooms by their number.
- 4- Sort guests by their number.
- 5- Search for all available rooms.
- 6- Search for guest by number.

Use three different sorting algorithms. Then calculate Big (O) of your program.

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Topic (3): College Management System:

College Management System: The required data is **Department** (ID& Name), **Course** (Code & Name & number of the student), and **Student** (ID& Name & GPA & Year & Phone).

- 1- Design a java program that takes the required data from user.
- 2- Sort Departments by their names.
- 3- Sort Courses by number of students.
- 4- Sort students by their GPA.
- 5- Search for department by its name.
- 6- Search for student by its ID.

Use three different sorting algorithms. Then calculate Big (O) of your program.

Topic (4): Club System

Club System: The required data is **Club** (Name, Branches, Manager, Location), **Members** (ID, Name, Phone, Number of children), and **Sports** (Name, ID, Number of teams).

- 1- Design a java program that takes the required data from user.
- 2- Sort Clubs by their names.
- 3- Sort Members by their IDs.
- 4- Sort Sports by their names.
- 5- Search for club by its name.
- 6- Search for member by its ID.

Use three different sorting algorithms. Then calculate Big (O) of your program.

General Rules:

- 1- Each team contains [3-5] members *from any class*.
- 2- You must use three different sorting algorithms and binary search.
- 3- You must to insert some data in the beginning of the program by calling functions in the main function.
- 4- Adding a GUI for your project is **Bonus**.
- 5- Project will be uploaded on Canvas, running code on your laptop and Printed report that describes the project. Report contains (Team members (ID, Name, Class) project name, Project Description contains brief comparison between algorithms used in the project).
- 6- Prepare well for the final discussion.

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